UNIT

I

ENGINEERING PHYSICS • DIGITAL NOTES

Wave Optics

Interference • Thin films • Newton's rings • Diffraction • Grating • Polarisation • Nicol prism • Wave plates

QUICK SUMMARY • What you will learn

- Why light waves add up to bright and dark fringes, and how thin films and Newton's rings use this
- How light bends around obstacles: single slit, double slit and the diffraction grating
- How light can be "combed" into one direction of vibration (polarisation) and where we use it



Fig. 1: Roadmap of Unit I: every topic follows from the wave nature of light

1 Interference of Light

Principle of superposition: when two or more waves meet at a point, the resultant displacement is the **vector sum** of the individual displacements. Waves pass through each other unchanged and separate again.

DEFINITION • Interference

The redistribution of light energy in space when two coherent waves superpose, giving alternate bright (maxima) and dark (minima) regions.

	Constructive interference	Destructive interference
Phase difference φ	0, 2π, 4π ... = 2nπ	π, 3π ... = (2n+1)π
Path difference Δ	nλ	(2n+1)λ/2
Resultant amplitude	$a_1 + a_2$ (maximum)	$\lvert a_1 - a_2 \rvert$ (minimum)
Intensity	$(a_1 + a_2)^2$	$(a_1 - a_2)^2$ (zero if $a_1 = a_2$)

Resultant intensity of two waves

$$I = I_1 + I_2 + 2\sqrt{I_1 I_2}\cos\phi$$

Phase ↔ path difference

$$\phi = \frac{2\pi}{\lambda}\Delta$$

REMEMBER • Conditions for sustained interference

- The sources must be **coherent** (constant phase difference, same frequency).
- Sources should be **monochromatic** and of nearly equal amplitude (for good contrast).
- Sources must be very close together/narrow; light is obtained from **one source split into two** (by division of wavefront or division of amplitude).
- Energy is conserved: the bright fringes gain exactly what the dark fringes lose.

Two ways of getting coherent sources

Method	Idea	Examples
Division of wavefront	Different parts of the same wavefront act as two sources	Young's double slit, Fresnel biprism
Division of amplitude	One beam is partly reflected and partly transmitted	Thin films, Newton's rings, Michelson interferometer

Young's fringe width (slit gap d, screen distance D)

$$\beta = \frac{\lambda D}{d}$$

2 Interference in Thin Films (Reflection Geometry)

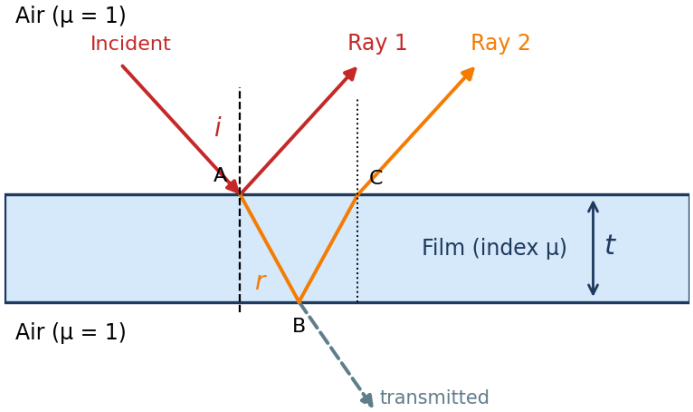


Fig. 2: A plane-parallel film of thickness t. Rays 1 and 2 come from the same incident ray and are coherent (division of amplitude).

Ray 1 reflects from the top surface (air → denser film), so it suffers a **phase change of π** (equivalent to λ/2). Ray 2 reflects at the bottom surface (film → air), where there is no phase change. Using geometry (Snell's law, μ = sin i / sin r):

Optical path difference (reflected)	$\Delta = 2\mu t\cos r \ \left(+\frac{\lambda}{2}\right)$
Bright film (maxima)	$2\mu t\cos r = (2n+1)\frac{\lambda}{2} \qquad n = 0, 1, 2, \ldots$
Dark film (minima)	$2\mu t\cos r = n\lambda \qquad n = 0, 1, 2, \ldots$

KEY IDEA • Transmitted light is complementary

In transmission, the conditions swap: bright when 2μt cos r = nλ and dark when 2μt cos r = (2n+1)λ/2. Where reflection shows a bright band, transmission shows a dark one.

Colours in thin films

- White light has many wavelengths. For a given thickness t and angle r, **only some wavelengths satisfy the bright condition**, so the film looks coloured (soap bubbles, oil on water, peacock feathers).
- Colour changes with thickness and viewing angle, because Δ = 2μt cos r changes.
- A very thin film (t ≪ λ) looks **black** in reflected light, as Δ ≈ λ/2 gives destructive interference (black spot on a bubble just before it bursts).
- A thick film shows no colours: too many orders overlap, so the eye sees white.

APPLICATIONS • Applications of thin-film interference

- **Anti-reflection (AR) coating:** a film of thickness t = λ/4μ and index μ ≈ $\sqrt{(\mu_{glass})}$ (MgF_2, μ = 1.38) cancels reflected light, so lenses, solar cells and spectacles transmit more light.
- **Highly reflecting coatings** (multilayer λ/4 stacks) for laser mirrors and dielectric mirrors.
- **Interference filters** that pass one narrow wavelength band.
- Testing the **flatness** of optical surfaces and measuring very small thicknesses.

EXAMPLE • Soap film

Find the minimum thickness of a soap film (μ = 1.33) that looks bright in reflected light of λ = 5890 Å at normal incidence.

2μt = λ/2 (n = 0) → t = λ/4μ = 5890×10^{-10} / (4 × 1.33) = **1.107×10^{-7} m (≈ 111 nm)**.

3 Newton's Rings

A plano-convex lens of large radius of curvature R is placed on an optically flat glass plate. A thin **air film of increasing thickness** forms between them. Monochromatic light falls normally (via a 45° glass plate G) and the interference of rays reflected from the top and bottom of the air film gives **concentric circular fringes** seen through the microscope M.

- Centre (t = 0): path difference is λ/2 (due to the π phase change), so the centre is **dark**.
- Rings get closer together as we move outward (t grows as r^2).
- Fringes are localised in the air film.

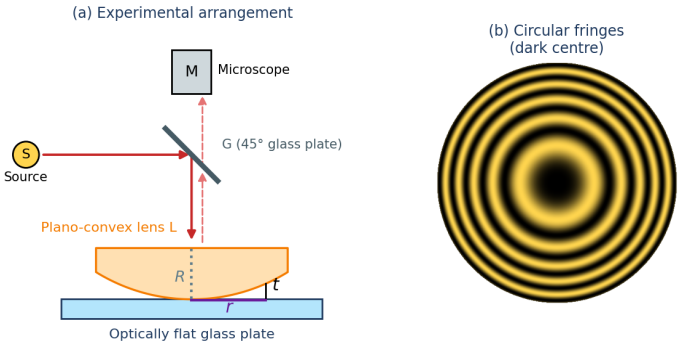


Fig. 3: Newton's rings set-up and pattern

Theory (reflected light, normal incidence, air film μ = 1)

For a ring of radius r at film thickness t: from the right triangle in the lens, $R^2 = r^2 + (R - t)^2 \rightarrow r^2 = 2Rt - t^2 \approx 2Rt$ (as $t \ll R$), hence $t = r^2/2R = D^2/8R$. The path difference is $2t + \lambda/2$.

Dark rings: $2t = n\lambda$	$D_n^2 = 4\,n\lambda R \qquad (D_n \propto \sqrt{n})$
Bright rings: $2t = (2n-1)\lambda/2$	$D_n^2 = 2\,(2n-1)\,\lambda R$
Wavelength of light	$\lambda = \dfrac{D_{n+p}^2 - D_n^2}{4\,p\,R}$
Refractive index of a liquid	$\mu = \dfrac{D_{n+p}^2 - D_n^2\ (\text{air})}{D_{n+p}^2 - D_n^2\ (\text{liquid})}$

KEY IDEA • Procedure

- **Wavelength:** measure the diameters of the n-th and (n+p)-th dark rings with a travelling microscope, using the known R; plot D^2 against n (straight line, slope = 4λR).
- **Refractive index:** fill the space between lens and plate with a liquid. The ring diameters shrink because the path difference becomes 2μt; $D^2 \rightarrow D^2/\mu$, so μ = (slope in air)/(slope in liquid).

EXAMPLE • Numerical

The diameters of the 4th and 12th dark rings are 0.400 cm and 0.700 cm. If λ = 5893 Å, find R.
$R = (D_{12}^2 - D_4^2) / (4\,p\,\lambda) = (4.9\times10^{-5} - 1.6\times10^{-5}) / (4 \times 8 \times 5.893\times10^{-7}) \approx$ **1.75 m**.

APPLICATIONS • Uses of Newton's rings

- Measuring the wavelength of monochromatic light and the refractive index of liquids.
- Finding the radius of curvature of a lens (R known → λ; λ known → R).
- Testing how flat or spherical a lens surface is: irregular rings show a defect.

REMEMBER • Common mistakes

- Use **diameter**, not radius, in $D^2 = 4n\lambda R$.
- Count rings from the **first dark ring** (n = 1); the centre is n = 0.
- Convert units: cm → m and Å → m before substituting.

4 Diffraction

DEFINITION • Diffraction

The bending of light around the edges of an obstacle or aperture and its spreading into the geometrical shadow. It is appreciable when the aperture size is comparable to λ.

Feature	Fresnel diffraction	Fraunhofer diffraction
Source / screen distance	Finite (near field)	Infinite (far field), usually using lenses
Wavefront	Spherical or cylindrical	Plane
Lenses needed?	No	Yes (collimating and focusing)
Analysis	Complicated (Fresnel zones)	Simple (Fourier-type)

Interference	Diffraction
Superposition of waves from two (or few) coherent sources	Superposition of secondary wavelets from one wavefront
All bright fringes are equally bright (ideal case)	Intensity falls rapidly for higher orders
Fringes usually equally spaced; dark fringes are nearly black	Fringes unequally spaced; minima are not perfectly dark

Fraunhofer diffraction at a single slit (width a)

The slit is divided into a large number of secondary sources; waves from different parts interfere. The path difference between waves from the two edges is a sinθ.

KEY IDEA • Why minima occur at a sinθ = nλ

- Split the slit into two halves. Every ray in the top half has a partner in the bottom half a distance a/2 away.
- If a sinθ = λ, the path difference of each pair is λ/2, so all pairs cancel and the intensity is zero. The same idea with 4, 6 ... parts gives nλ.
- Maxima lie roughly midway: a sinθ ≈ (2n+1)λ/2.

Intensity	$I = I_0\left(\frac{\sin\alpha}{\alpha}\right)^2, \qquad \alpha = \frac{\pi a \sin\theta}{\lambda}$
Minima (α = nπ)	$a\sin\theta = n\lambda, \quad n = \pm 1, \pm 2, \ldots$
Secondary maxima	$\tan\alpha = \alpha \;\; (\alpha \approx 1.43\pi,\ 2.46\pi, \ldots)$

APPLICATIONS • Applications of diffraction

- Grating spectrometers to study the spectra of stars and materials.
- X-ray diffraction to find crystal structure (Unit II).
- CD/DVD surfaces show rainbow colours; limits the resolving power of telescopes and microscopes.

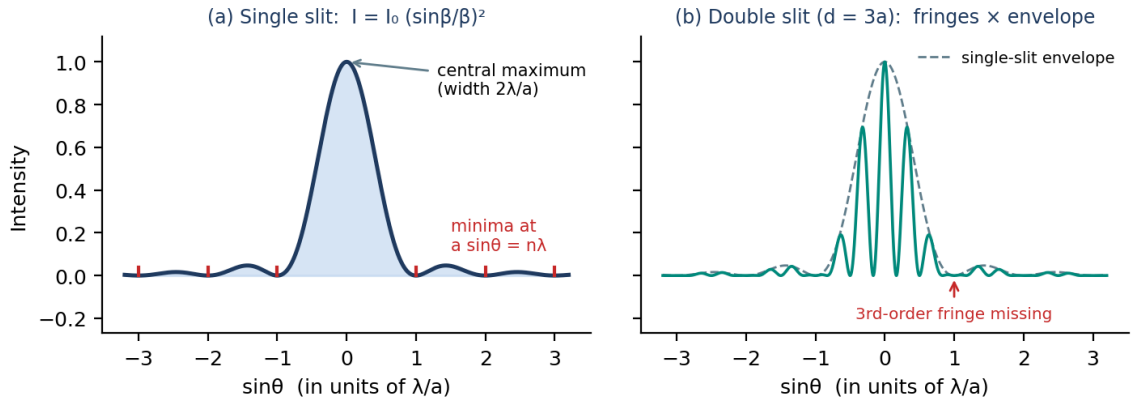


Fig. 4: (a) single-slit pattern, (b) double-slit pattern = cos² fringes inside a single-slit envelope

- Central maximum is the brightest and twice as wide as the others (angular width 2λ/a); secondary maxima fall as 1 : 0.045 : 0.016 ...
- Narrower slit (smaller a) gives a **wider** pattern.

5 Double Slit, N-Slits and the Diffraction Grating

Double slit (slit width a, separation d): the intensity is the product of the single-slit diffraction envelope and the two-source interference term.

Double slit intensity	$I = I_0\left(\frac{\sin\alpha}{\alpha}\right)^2\cos^2\beta, \qquad \beta = \frac{\pi d\sin\theta}{\lambda}$
Interference maxima	$d\sin\theta = n\lambda$
Missing orders if d/a = m	$n = m,\ 2m,\ 3m, \ldots$

N slits (qualitative): the pattern gets an extra factor (sin Nβ / sin β)². Principal maxima stay at d sinθ = nλ but become **sharper and brighter (intensity ∝ N²)**. Between two principal maxima there are **(N−1) minima** and **(N−2) weak secondary maxima**.

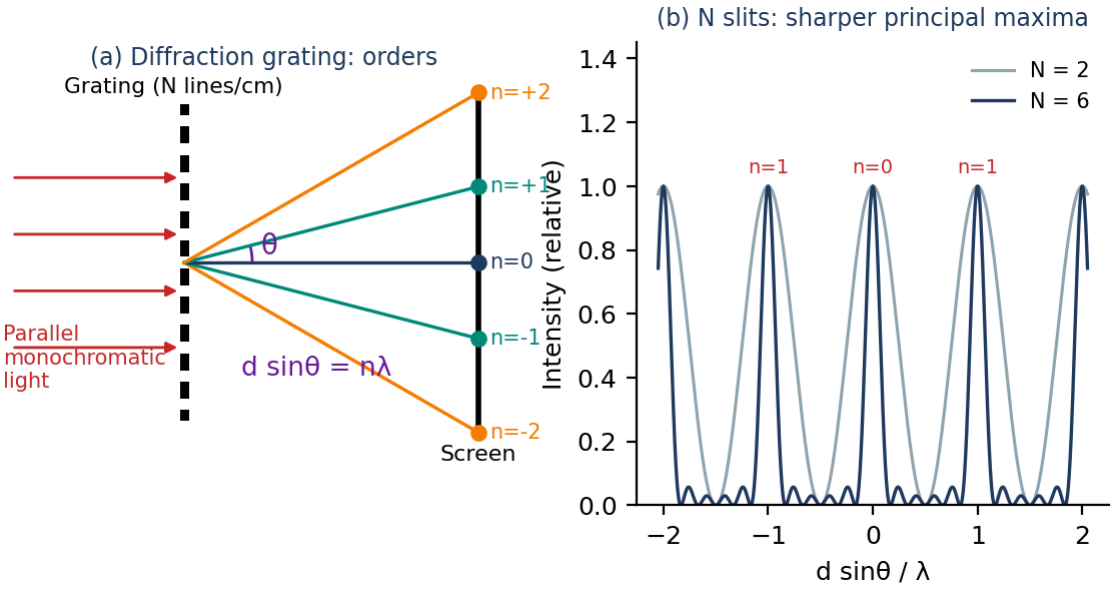


Fig. 5: (a) orders from a grating; (b) more slits → narrower principal maxima

DEFINITION • Diffraction grating

A glass or metal plate with a very large number of equally spaced, parallel, narrow slits (about 6000 lines/cm). Grating element (a + b) = 1/N, with a = slit width and b = opaque width.

Grating equation	$(a+b)\sin\theta_n = n\lambda$
Highest possible order	$n_{max} < \frac{a+b}{\lambda}$

Dispersive power and resolving power (qualitative)

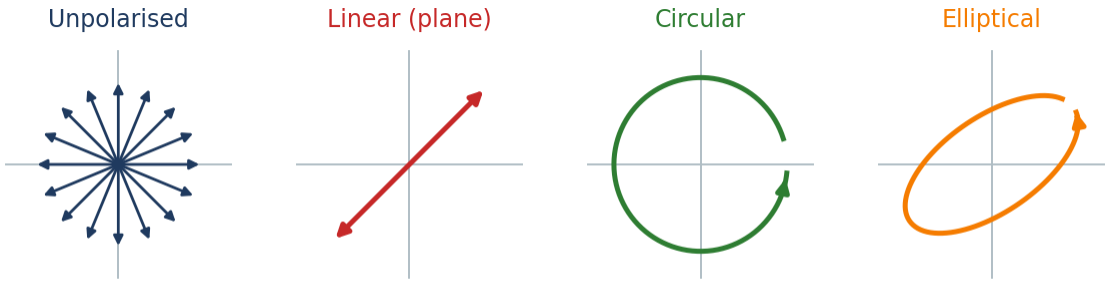
Quantity	Meaning	Expression	How to increase
Dispersive power (angular dispersion)	Angular separation dθ of two wavelengths differing by dλ	dθ/dλ = n / [(a+b) cosθ]	Higher order n, finer grating (small a+b)
Resolving power	Ability to separate two close spectral lines (Rayleigh criterion)	R = λ/dλ = nN	Higher order n, more lines N illuminated

EXAMPLE • Grating numerical

A grating has 6000 lines/cm. For λ = 5000 Å: a + b = 1/6000 cm = 1.667×10^{-6} m, so n_{max} = $1.667\times10^{-6}/5\times10^{-7}$ = 3.3, i.e. **3 orders**. For n = 2: sinθ = 2λ/(a+b) = 0.6, so θ = **36.9°**.

6 Polarisation

Light is a **transverse** electromagnetic wave. In ordinary (unpolarised) light the electric vector vibrates in all directions perpendicular to the ray. **Polarisation** restricts the vibrations to a single direction or a definite pattern. Longitudinal waves (e.g. sound) cannot be polarised.



Tip of the electric vector as seen looking against the beam (E_x–E_y plane)

Fig. 6: Types of polarisation

Type	Description	Produced by
Plane (linear)	E vibrates along one line	Polariser (Nicol, Polaroid)
Circular	Two perpendicular components, equal amplitude, phase difference π/2; tip traces a circle	Plane light at 45° through quarter-wave plate
Elliptical	Unequal amplitudes and/or phase difference other than 0, π/2, π	Plane light through a wave plate at an angle other than 45°
Partial	Mixture of polarised and unpolarised light	Reflection, scattering

Malus's law (analyser at angle θ) $$I = I_0 \cos^2\theta$$

Methods of producing plane polarised light

(i) Reflection (Brewster's law). At the particular angle of incidence i_B the reflected light is completely plane polarised (vibrations ⊥ plane of incidence) and the reflected and refracted rays are at 90°.

Brewster's law $$\tan i_B = \mu \qquad (i_B + r = 90^\circ)$$

Example: glass (μ = 1.5) → $i_B = \tan^{-1}1.5 = 56.3°$; water (μ = 1.33) → 53.1°.

(ii) Refraction (pile of plates). At i_B each glass plate reflects only the ⊥ component, so the transmitted beam gets richer in the parallel component. With about 15–20 plates the transmitted light is nearly plane polarised.

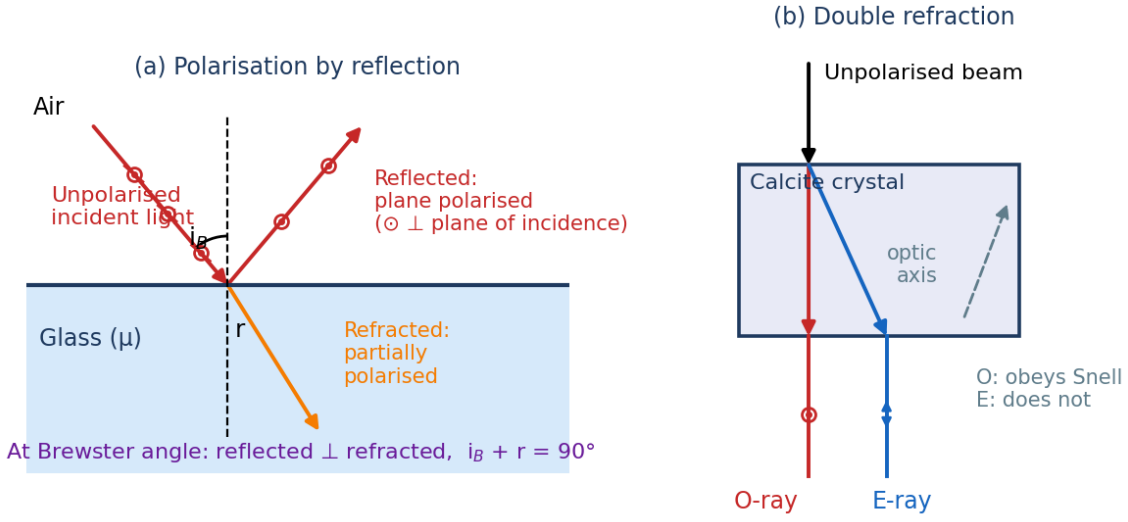


Fig. 7: (a) polarisation by reflection at the Brewster angle; (b) double refraction in calcite

7 Double Refraction and Nicol Prism

(iii) Double refraction (birefringence). A calcite or quartz crystal splits one ray into two:

- **Ordinary (O) ray:** obeys Snell's law, same speed in all directions, vibrations ⊥ principal section.
- **Extraordinary (E) ray:** does not obey Snell's law, speed varies with direction, vibrations in the principal section.
- Both are plane polarised at right angles to each other. Along the **optic axis** there is no double refraction.

- **Negative crystal** (calcite): $\mu_o > \mu_e$ (1.658 and 1.486). **Positive crystal** (quartz): $\mu_e > \mu_o$ (1.553 and 1.544).

Nicol prism

A calcite crystal (length : width = 3 : 1) with its end faces ground from 71° to 68°, cut along the diagonal and cemented with **Canada balsam** (μ = 1.55).

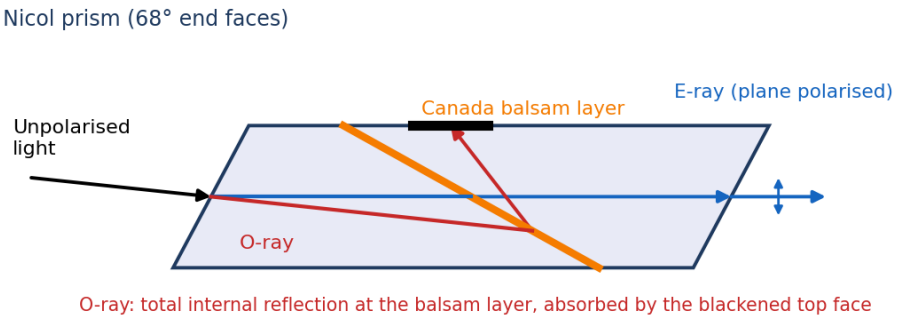


Fig. 8: Working of a Nicol prism

- Balsam is **rarer than calcite for the O-ray** (1.658 > 1.55) and denser for the E-ray (1.486 < 1.55).
- The O-ray meets the balsam at an angle greater than the critical angle (≈ 69°), so it undergoes **total internal reflection** and is absorbed by the blackened side.
- The E-ray is transmitted, so the emergent light is **plane polarised**. Used as a **polariser**, and a second Nicol acts as an **analyser**.

REMEMBER • Limitations
Limited field of view (about 29°), costly, and the emergent beam is slightly displaced laterally. Polaroid sheets are cheaper alternatives.

8 Half-Wave and Quarter-Wave Plates

A thin slice of a uniaxial crystal cut **parallel to its optic axis**. Light entering normally splits into O and E rays that travel along the same path at different speeds, so a phase difference builds up between them.

Path difference	$\Delta = (\mu_o - \mu_e)\, t$
Quarter-wave plate ($\Delta = \lambda/4$)	$t = \dfrac{\lambda}{4\,\lvert\mu_o - \mu_e\rvert}$
Half-wave plate ($\Delta = \lambda/2$)	$t = \dfrac{\lambda}{2\,\lvert\mu_o - \mu_e\rvert}$

	Quarter-wave plate (QWP)	Half-wave plate (HWP)
Phase difference	π/2 (path λ/4)	π (path λ/2)
Plane light at 45° to axis	Becomes **circularly** polarised	Stays plane, rotated by 90°
Plane light at other angle θ	Becomes elliptically polarised	Plane, rotated by 2θ
Reverse use	Circular → plane	Flips handedness of circular light



Fig. 9: Producing circularly polarised light

EXAMPLE • Brewster and Malus
A glass slab (μ = 1.5) is used at the polarising angle, so $i_B = \tan^{-1}(1.5) = 56.3°$ and r = 33.7°. Plane polarised light of intensity I_0 passes through an analyser at 60°: $I = I_0\cos^2 60° = \mathbf{I_0/4}$.

EXAMPLE • Quartz QWP
Quartz: $\mu_e = 1.553$, $\mu_o = 1.544$, $\lambda = 5890$ Å. $t = \lambda / 4(\mu_e - \mu_o) = 5.89\times10^{-7} / (4 \times 0.009) =$ **16.4 µm**.

APPLICATIONS • Applications of polarisation

- Polaroid sunglasses and camera filters cut reflected glare.
- LCD/LED displays, 3-D cinema (circular polarisers).
- Stress analysis (photoelasticity) in transparent models.
- Optical activity measurement in sugar (polarimeter); optical isolators in lasers and fibre links.

More solved problems

EXAMPLE • AR coating

Find the thickness of a MgF_2 film ($\mu = 1.38$) that makes a lens non-reflecting at $\lambda = 5500$ Å: $t = \lambda/4\mu = 5.5\times10^{-7}/(4 \times 1.38) =$ **99.6 nm**.

EXAMPLE • Resolving power

A grating with 10^4 lines illuminated fully is used in the 2nd order. $R = nN = 2 \times 10^4$. Near $\lambda = 5890$ Å, $d\lambda = \lambda/R = 5890/20000 =$ **0.29 Å**, so the sodium D lines (6 Å apart) are easily resolved.

EXAMPLE • Newton's rings in liquid

In air $D_{10}^2 - D_5^2 = 0.0145$ cm²; with a liquid it is 0.0109 cm². $\mu = 0.0145/0.0109 =$ **1.33** (water).

Key terms at a glance

Term	Meaning
Coherent sources	Sources with a constant phase difference and the same frequency
Fringe width	Distance between two consecutive bright (or dark) fringes
Optical path	Refractive index × geometrical path ($\mu \times x$)
Order (n)	Integer labelling the maxima or minima, n = 0, 1, 2 ...
Rayleigh criterion	Two lines are just resolved when the maximum of one falls on the first minimum of the other
Principal section	Plane through the optic axis and perpendicular to two opposite faces of the crystal
Analyser	Second polariser used to detect or analyse polarised light

KEY IDEA • Real-life connections

- The colours of a CD, a soap bubble and a butterfly wing are all interference or diffraction effects.
- Camera lenses look purple-green because the AR coating cancels yellow-green reflections best.
- Polaroid sunglasses block horizontally polarised glare reflected from roads and water.

9 Formula Sheet and Exam Corner

Topic	Key formulae
Thin film (reflected)	$2\mu t \cos r = (2n+1)\lambda/2$ (bright); $= n\lambda$ (dark). AR coating: $t = \lambda/4\mu$
Newton's rings	$D_n^2 = 4n\lambda R$ (dark); $\lambda = (D_{n+p}^2 - D_n^2)/4pR$; μ = slope(air)/slope(liquid)
Single slit	$a \sin\theta = n\lambda$ (minima); $I = I_0 (\sin\alpha/\alpha)^2$
Grating	$(a+b) \sin\theta = n\lambda$; $d\theta/d\lambda = n/[(a+b)\cos\theta]$; $R = nN$
Polarisation	Malus: $I = I_0\cos^2\theta$; Brewster: $\tan i_B = \mu$
Wave plates	QWP: $t = \lambda/4\lvert\mu_o - \mu_e\rvert$; HWP: $t = \lambda/2\lvert\mu_o - \mu_e\rvert$

EXAM CORNER • Short questions (2–3 marks)

- State the principle of superposition and the conditions for sustained interference.
- Why is the centre of Newton's rings dark in reflected light?
- Differentiate between Fresnel and Fraunhofer diffraction.
- What is a diffraction grating? Write the grating equation.
- State Brewster's law and Malus's law.
- Define the O-ray, E-ray and optic axis.
- Why do thin films show colours in white light?

EXAM CORNER • Long questions (10 marks)

- Derive the conditions for bright and dark fringes in a thin film in reflected light. Explain colours in thin films.
- Explain the formation of Newton’s rings. Derive the expression for the diameter of dark rings and describe how the wavelength is determined.
- Describe Fraunhofer diffraction at a single slit and obtain the intensity distribution with a neat diagram.
- Explain the construction and working of a Nicol prism.
- Explain half-wave and quarter-wave plates, with the thickness expressions.
- Write notes on the dispersive power and resolving power of a grating.

EXAM CORNER • Quick MCQs (answers at the end of the box)

- 1. The centre of Newton’s rings in reflected light is (a) bright (b) dark (c) coloured.
- 2. At the Brewster angle the angle between reflected and refracted rays is (a) 0° (b) 45° (c) 90°.
- 3. Resolving power of a grating with N lines in order n is (a) n/N (b) nN (c) N/n.
- 4. A quarter-wave plate turns 45° plane polarised light into (a) circular (b) unpolarised (c) rotated plane light.
- **Answers:** 1-b, 2-c, 3-b, 4-a.

QUICK SUMMARY

- Interference = redistribution of energy; needs coherent sources. Thin film: $\Delta = 2\mu t \cos r$ with an extra $\lambda/2$ in reflection.
- Diffraction = bending around apertures; grating gives sharp orders with high dispersive and resolving power.
- Polarisation proves light is transverse; it is produced by reflection, refraction, double refraction and wave plates.